

## SAFETY DATA SHEET

Revision Date 30-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** cobalt perchlorate, hexahydrate reagent

**Cat No. :** 11652

**CAS No** 13478-33-6  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |             |
|--------------------------------------------------|-------------|
| Oxidizing solids                                 | Category 2  |
| Skin Corrosion/Irritation                        | Category 2  |
| Serious Eye Damage/Eye Irritation                | Category 2  |
| Respiratory Sensitization                        | Category 1  |
| Carcinogenicity                                  | Category 1B |
| Specific target organ toxicity (single exposure) | Category 3  |
| Target Organs - Respiratory system.              |             |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May intensify fire; oxidizer

Causes skin irritation  
Causes serious eye irritation  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
May cause respiratory irritation  
May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Wear eye/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
In case of inadequate ventilation wear respiratory protection  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep/Store away from clothing/ other combustible materials  
Take any precaution to avoid mixing with combustibles

**Response**

IF exposed or concerned: Get medical attention/advice

**Inhalation**

If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician  
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                               | CAS No     | Weight % |
|-----------------------------------------|------------|----------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | <=100    |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                           |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                  |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                     |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                            |

## 5. Fire-fighting measures

|                                         |                                                                            |
|-----------------------------------------|----------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Foam. Water may be ineffective. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                   |
| <b>Flash Point</b>                      | No information available                                                   |
| <b>Method -</b>                         | No information available                                                   |
| <b>Autoignition Temperature</b>         | No information available                                                   |
| <b>Explosion Limits</b>                 |                                                                            |
| <b>Upper</b>                            | No data available                                                          |
| <b>Lower</b>                            | No data available                                                          |
| <b>Oxidizing Properties</b>             | Oxidizer                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                   |

### Specific Hazards Arising from the Chemical

Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

### Hazardous Combustion Products

Hydrogen chloride. Cobalt oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
- OX

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                                      |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Sweep up and shovel into                                           |

suitable containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from clothing and other combustible materials.

**Storage.** Store under an inert atmosphere. Protect from moisture. Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Incompatible Materials. Reducing Agent. Strong reducing agents. Combustible material.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                               | ACGIH TLV                   | OSHA PEL | NIOSH | Mexico OEL (TWA)            |
|-----------------------------------------|-----------------------------|----------|-------|-----------------------------|
| cobalt perchlorate, hexahydrate reagent | TWA: 0.02 mg/m <sup>3</sup> |          |       | TWA: 0.02 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | Solid Crystalline        |
| Appearance                       | Red                      |
| Odor                             | Odorless                 |
| Odor Threshold                   | No information available |
| pH                               | No information available |
| Melting Point/Range              | 1534 °C / 2793.2 °F      |
| Boiling Point/Range              | No information available |
| Flash Point                      | No information available |
| Evaporation Rate                 | Not applicable           |
| Flammability (solid,gas)         | No information available |
| Flammability or explosive limits |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |
| Vapor Pressure                   | No information available |
| Vapor Density                    | Not applicable           |
| Specific Gravity                 | 3.3 g/cm <sup>3</sup>    |

|                                               |                                                        |
|-----------------------------------------------|--------------------------------------------------------|
| <b>Solubility</b>                             | No information available                               |
| <b>Partition coefficient; n-octanol/water</b> | No data available                                      |
| <b>Autoignition Temperature</b>               | No information available                               |
| <b>Decomposition Temperature</b>              | No information available                               |
| <b>Viscosity</b>                              | Not applicable                                         |
| <b>Molecular Formula</b>                      | Co(Cl O <sub>4</sub> ) <sub>2</sub> .6H <sub>2</sub> O |
| <b>Molecular Weight</b>                       | 366.74 (257.83anhy)                                    |

## 10. Stability and reactivity

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                       |
| <b>Stability</b>                        | Hygroscopic. Oxidizer: Contact with combustible/organic material may cause fire.          |
| <b>Conditions to Avoid</b>              | Exposure to moist air or water. Incompatible products. Excess heat. Combustible material. |
| <b>Incompatible Materials</b>           | Reducing Agent, Strong reducing agents, Combustible material                              |
| <b>Hazardous Decomposition Products</b> | Hydrogen chloride, Cobalt oxides                                                          |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                             |                          |
|---------------------------------------------|--------------------------|
| <b>Toxicologically Synergistic Products</b> | No information available |
|---------------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                               | CAS No     | IARC       | NTP                    | ACGIH | OSHA       | Mexico     |
|-----------------------------------------|------------|------------|------------------------|-------|------------|------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | Not listed | Reasonably Anticipated | A3    | Not listed | Not listed |

NTP: (National Toxicity Program)

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

|                                 |                           |
|---------------------------------|---------------------------|
| <b>Mutagenic Effects</b>        | No information available  |
| <b>Reproductive Effects</b>     | No information available. |
| <b>Developmental Effects</b>    | No information available. |
| <b>Teratogenicity</b>           | No information available. |
| <b>STOT - single exposure</b>   | Respiratory system        |
| <b>STOT - repeated exposure</b> | None known                |

|                                                   |                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                              |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                              |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                        |

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | based on information available. May persist                           |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                                      |
|-----------------------------|--------------------------------------|
| <b>UN-No</b>                | UN1481                               |
| <b>Proper Shipping Name</b> | PERCHLORATES, INORGANIC, N.O.S.      |
| <b>Technical Name</b>       | (Cobalt(II) perchlorate hexahydrate) |
| <b>Hazard Class</b>         | 5.1                                  |
| <b>Packing Group</b>        | II                                   |

### TDG

|                             |                                 |
|-----------------------------|---------------------------------|
| <b>UN-No</b>                | UN1481                          |
| <b>Proper Shipping Name</b> | PERCHLORATES, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 5.1                             |
| <b>Packing Group</b>        | II                              |

### IATA

|                             |                                 |
|-----------------------------|---------------------------------|
| <b>UN-No</b>                | UN1481                          |
| <b>Proper Shipping Name</b> | PERCHLORATES, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 5.1                             |
| <b>Packing Group</b>        | II                              |

### IMDG/IMO

|                             |                                 |
|-----------------------------|---------------------------------|
| <b>UN-No</b>                | UN1481                          |
| <b>Proper Shipping Name</b> | PERCHLORATES, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 5.1                             |
| <b>Packing Group</b>        | II                              |

## 15. Regulatory information

### United States of America Inventory

| Component                               | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | -    | -                                             | -                           |

### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                               | CAS No     | DSL | NDL | EINECS | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-----------------------------------------|------------|-----|-----|--------|-------|------|------|------|-------|------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | -   | -   | -      | -     | -    |      | -    | -     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component                               | CAS No     | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-----------------------------------------|------------|----------|-------------------------------|---------------------------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | <=100    | 0.1 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

| Component                               | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-----------------------------------------|-----------|-------------------------|-------------------------|
| cobalt perchlorate, hexahydrate reagent | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                               | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------------------------|---------------|------------|--------------|----------|--------------|
| cobalt perchlorate, hexahydrate reagent | -             | X          | X            | X        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

#### Other International Regulations

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component                               | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | -                                                                   | -                                                                             | -                                                                                                     |

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                               | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**  
 Not applicable

#### Other International Regulations

| Component                               | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| cobalt perchlorate, hexahydrate reagent | 13478-33-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

### 16. Other information

**Prepared By** Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

**Revision Date** 30-Mar-2024  
**Print Date** 30-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other

materials or in any process, unless specified in the text

**End of SDS**